GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Decommissioning Plan
Document Number	EA-BFPCL-GPMC-GGPL-FEED-OPS-PLN-002
Document Revision	R01
Document Status	Issued for Review
Originator	Olarinde Ige
Security Classification	Restricted
Issue Date	13-May-22

Decommissioning Plan

R01	13.05.22	Issued for Review	O.Ige	M.Ekpunobi	K.Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

R01	13.05.22	Issued for Review	O.Ige	M.Ekpunobi	K.Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Review Code & Status (to be filled by Client)		
☐ Code 1	**Approved** – Further work may proceed.	
☐ Code 2	**Approved with comments.** – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed	
☐ Code 3	**Reviewed** – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed	
☐ Code 4	**Retained** for information only	
☐ Code 5	**Review Not Required** – Do not Resubmit	
Date:	Position:	Sign:

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-OPS-PLN-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Project Information, Terms, Abbreviations, and Definitions

Project Information – General

1.0	Employer	Brass Fertilizer & Petrochemical Company Limited (BFPCL)
2.0	PMC	Tata Consulting Engineers Limited (TCE)
3.0	EPCC Contractor	China Tianchen Engineering Corporation (TCC)
4.0	Project Title	Gas Processing and Methanol Complex (GPMC)
5.0	Location	Akabel, Brass Island: • 85km South West of Port Harcourt, • 70km South of Yenagoa and • 90km West of Bonny Island.
5.1	State	Bayelsa State, Nigeria
5.2	Nearest Air Port, Sea Port. Railway Station	Port Harcourt (PHC), 85km Bayelsa Airport (Under Construction)
6.0	Standard Process Condition	15°c and 1.01325 bar
7.0	Design Life	30 years

Definition of Terms

Terms	Definitions
BFPCL	Brass Fertilizer & Petrochemical Limited.
SPDC	Shell Petroleum Development Company Nig. Limited
Employer	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase1)

Abbreviations

Abbreviation	Meaning
AG	Associated Gas
ANSI	American National Standards Institute
API	American Petroleum Institute
BBOU	Bubouwe Bou Manifold
BFPCL	Brass Fertilizer & Petrochemical Company Limited
BLV	Blowdown Valve
CAO	Computer Aided Operations
ESD	Emergency Shut Down
FEED	Front End Engineering Design
FLB	Field Logistics Base
FLM	First Line Maintenance
FQI	Flow Quantity Indicator
GSPA	Gas Sales & Purchase Agreement
HP	High Pressure
HSSE	Health Safety Security and Environment
LCP	Local Control Panel
LCV	Level Control Valve
LP	Low Pressure
MSDS	Material Safety Data Sheet
MTPA	Metric Tonnes per Annum
MTPD	Metric Tonnes Per Day
NAG	Non-Associated Gas
NUPRC	Nigerian Upstream Petroleum Regulatory Commission
OML	Operating Mining License
OSD	Operational Shut Down

PAS	Process Automation System
PC	Pressure Controller
PCV	Pressure Control Valve
PEFS	Process Engineering Flow Scheme
PTW	Permit to Work
SPDC	Shell Petroleum Development Company of Nigeria

Site and Environmental Conditions*

	Air Temperature	Average daytime 32oC; rainy season 28oC; night time 23oC
	Weather	Dry Season (Harmattan); November – January; the atmosphere is laden with extremely fine dust (5 microns particle size) Wet Season: heavy rainfall, two peaks, August- 245mm, September-331mm
	Relative Humidity	Mean 81%; Minimum 69% & Maximum 88%
	Wind Speed and Direction	monthly mean 8.5km/h; varies from 6.2km/h to 9.6km/h The south-westerly wind is 60% predominant during the wet season (March - October). The north-easterly wind is 32% predominant during the dry season (November– February)
	Seismicity	The project area is classified as Seismic Zone 0

*Data from ESIA Report

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel, - as shown in Figures 1.1, 1.2, & 1.3.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

Gas Gathering Concept, Phase-1 (BBOU & ELEP): E-MF at ELEP and S-MF at BBOU

Nodal Gas Gathering at BBOU & ELEP with BBOU Node Gas Joining ELEP Node for Onward Transportation to the Gas Processing Plant at AKAB

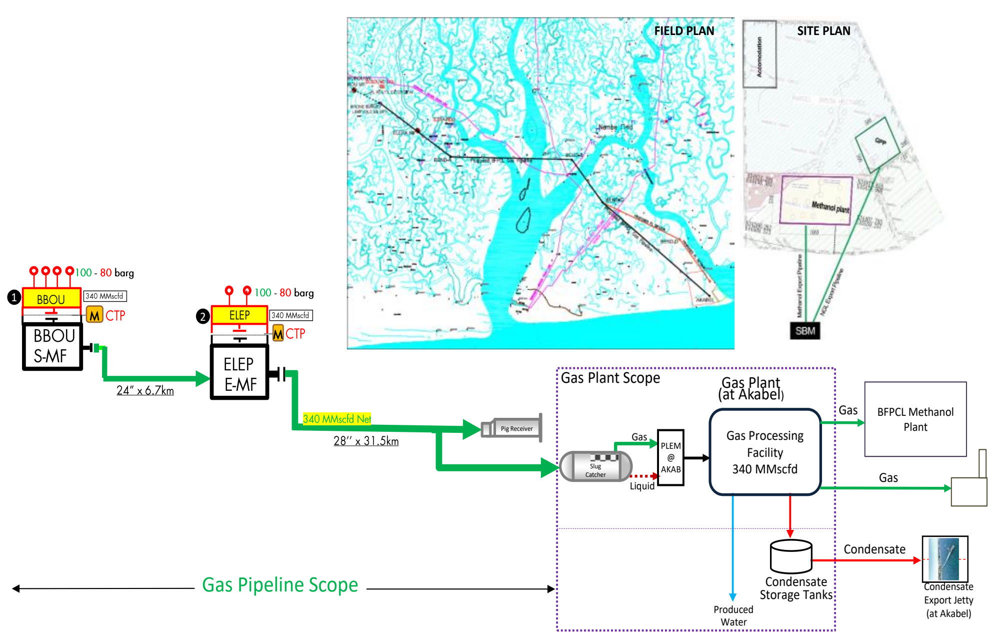

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this Decommissioning Plan is to provide a guideline to ensure that the gas gathering pipelines and the associated infrastructure is safely decommissioned.

This Decommissioning Plan for Brass Fertilizer Petrochemical Company Limited (BFPCL) Gas Gathering Pipelines and Associated Infrastructure FEED defines the activities to be carried out for safe decommissioning of the pipelines at the end of the life cycle.

This document sets out how the pipelines and manifolds facilities will be decommissioned at the end of its full life cycle safely and in compliance with the environmental laws of Nigeria.

This document shall be updated during the detailed engineering phase of this project.

A separate document will be developed for the decommissioning of the Methanol Complex outside this contract

The FEED shall fully comply with all specified relevant contractual requirements.

1.3. Scope

The Facilities to be covered under the project scope are:

- Satellite Manifold at BBOU complete with all Process, Utility & Control Infrastructure, however with provision for PIG receiver in phase 2 production from OPUG.
- Evacuation Manifold at ELEP complete with all Process, Utility & Control Infrastructure, however with provision for PIG receiver in phase 2 production from NUNR.
- Pipeline from BBOU to ELEP
- Pipeline from ELEP to AKAB (up to Inlet Flange of Slug Catcher)
- PIG Launchers at BBOU and ELEP Manifold
- PIG Receivers at ELEP and AKAB
- Manifold control system also to be provided. Control Room required at each Manifold to house the Manifold Control system & Telecom system
- Cabinet Room to house the electrical MCC.
- UPS Power for Control System, associated Civil, Mechanical, Pipeline, Electrical, Instrumentation & Control, Telecommunication, and all other supporting infrastructure.

The Scope of the FEED Engineering covers the provision of all necessary information, technical drawings, philosophies, specifications, design data, listing of codes and standards required for the detailed design and construction of the Multiphase pipeline system.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendment or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1. Concept Engineering Referenced Documents

Document Number	Document Title

2.2. Shell Design and Engineering Practices (DEPs)

Applicable DEPs, supplements and MESC SPE Codes are as referenced in various discipline dossiers, some of which are listed below.

Document Number	Title

2.3. International Codes And Standard

Discipline	Document Number	Title

3. General Design Data

3.1. Design Units of Measurement

Variable	Unit

4. Pipeline Deactivation and Abandonment

In the operational phase of a pipeline, it may be required to deactivate or abandon a pipeline for various reasons including reservoir decline. In such a situation, safety of personnel and environmental considerations shall be paramount.

The pipelines shall be isolated from all process sources and the pipelines flushed to remove any resident hydrocarbons. The flushed contents shall be collected in drums for safe disposal in line with regulatory requirements.

4.1. Pipeline Deactivation

Pipeline deactivation refers to the cessation of use of a pipeline for some period of time. In view of the fact that a pipeline is a valuable piece of property, a deactivated pipeline shall be maintained so that it may be restored to service in the nearest future. Any pipeline that has not had fluid flow for 18 consecutive months must be deactivated.

The following are very key for pipeline deactivation.

- Decommission the pipeline by mechanical pigging and water treated with oxygen scavengers, biocide and chemical/corrosion inhibitors.
- Decommission non-piggable or difficult to pig pipelines by enhanced chemical cleaning.
- All debris and effluent recovered from the pipeline must be disposed/evacuated in an environmentally friendly manner as approved by the Company.
- Deactivated line external coating condition must be ascertained for repairs of defective areas and prevent external corrosion while abandoned.
- Deactivated line must be installed with sacrificial anodes for external corrosion control over the period of abandonment.
- Injection plug must be installed on the deactivated pipeline (if none already exists on the pipeline) for the purpose of injecting the preservative fluids and must be corked or plugged or sealed before finally abandoning the pipeline.

The following minimum information must be made available to all stakeholders when a pipeline is deactivated.

- Reason for deactivation of the pipeline
- Method of isolation
- Pressure left on the pipeline
- The medium used to fill the pipeline and the effects of the medium on the integrity of the pipeline
- Method being used for internal and external corrosion monitoring and mitigation
- Planned length of deactivation

- Planned maintenance activities on the pipeline during the deactivation time frame.

Customer supply connections shall be closed and locked or disconnected.

Pipeline segments shall be physically disconnected from all sources and supplies of product, including connecting pipelines, crossover piping, meter station control piping, and other apparatus.

All disconnected piping ends or other openings shall be sealed with blind flanges, welded caps, or other sealing devices as appropriate for the connection.

Pipeline markers and casing vents shall not be removed.

However, if the pipeline will be resuming production, an engineering assessment is required, and a pressure test may be required as part of the assessment.

4.2. Pipeline Abandonment

Operator must disconnect both ends, purge, and seal each end before abandonment or a period of deactivation where the pipeline is not being maintained.

Abandoned Oil pipelines must be filled with water added with Oxygen scavenger, biocide and Corrosion inhibitors. Abandoned Gas lines must be filled with inert gas (nitrogen, Helium, etc).

Each inactive pipeline that is not being maintained must be disconnected from all hydrocarbon sources/supplies, purged, and sealed at each end.

Whenever service to a customer is discontinued, the following procedures must be adopted:

- The valve that is closed to prevent the flow of hydrocarbon to the customer must be provided with a locking device or other means designed to prevent the opening of the valve by persons other than those authorized by the operator.
- A mechanical device or fitting that will prevent the flow of hydrocarbon must be installed in the service line or in the meter assembly.
- The customer's piping must be physically disconnected from the hydrocarbon supply and the open pipe ends sealed.

All above ground elements and components shall be removed.

ROW owners shall be notified that the line will no longer be operated.

Plan for pipeline abandonment shall include updating NUPRC and shall meet regulatory requirements before the pipeline is abandoned.

5. Health, Safety and Environment (HSE)

- Contractors providing services for the decommissioning of the pipelines shall develop HSE plans, setting out performance targets and methods to be adopted to achieve set targets.
- All activities shall be carried out in compliance with BFPCL HSE policy. The objective will be to assure that the health of employees, contractor personnel and others working on the decommissioning activities will not be adversely affected by the work.
- A PTW (Permit to Work) system shall be implemented for all decommissioning activities since these are non- routine work.
- Special precautions shall be taken when hot work is required for pipe breaking, etc. since the pipelines previously contained hydrocarbons
- If required for preservation, selection, approval, and procurement of chemicals will be in accordance with BFPCL approved procedure. Chemicals must be acquired, transported, used, and disposed of in compliance with NUPRC, NMDPRA and FME regulations.
- All Chemicals will be accompanied with the appropriate Material Safety Data Sheet (MSDS).
- Environmental Impact Assessments (EIA) shall be conducted post decommissioning in order to evaluate impact of the operation on the environment.